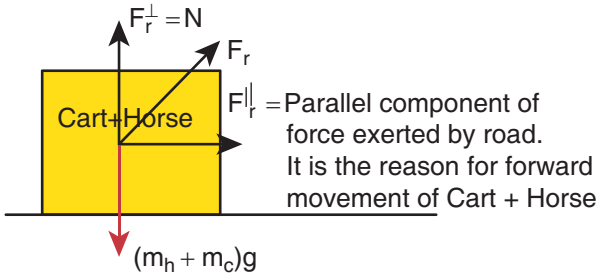
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If we take the cart+horse as a system, then there are two forces acting on the system.

(i) Downward gravitational force $(m_h + m_c)g$

(ii) The force exerted by the road (F_r) on the system.

It is shown in the following figure.



(iii) In this case the force exerted by the road (F_r) on the system (cart+horse) is resolved in to parallel and perpendicular components. The perpendicular component is the normal force which cancels the downward gravitational force $(m_h + m_c)g$. The parallel component of the force is not balanced, hence the system (cart+horse) accelerates and moves forward due to this force.

EXAMPLE 3.13

The position of the particle is represented by $y = ut - \frac{1}{2}gt^2$.

a) What is the force acting on the particle?
b) What is the momentum of the particle?

Solution

To find the force, we need to find the acceleration experienced by the particle.

The acceleration is given by $a = \frac{d^2y}{dt^2}$

$$\text{(or)} \quad a = \frac{dv}{dt}$$

Here

v = velocity of the particle in y direction

$$v = \frac{dy}{dt} = u - gt$$

The momentum of the particle = mv = m (u-gt).

$$a = \frac{dv}{dt} = -g$$

The force acting on the object is given by $F = ma = -mg$

The negative sign implies that the force is acting on the negative y direction. This is exactly the force that acts on the object in projectile motion.

3.3.2 Particle Moving in an Inclined Plane

When an object of mass m slides on a frictionless surface inclined at an angle θ as shown in the Figure 3.12, the forces acting on it decides the

a) acceleration of the object
b) speed of the object when it reaches the bottom

The force acting on the object is

(i) Downward gravitational force (mg)
(ii) Normal force perpendicular to inclined surface (N)